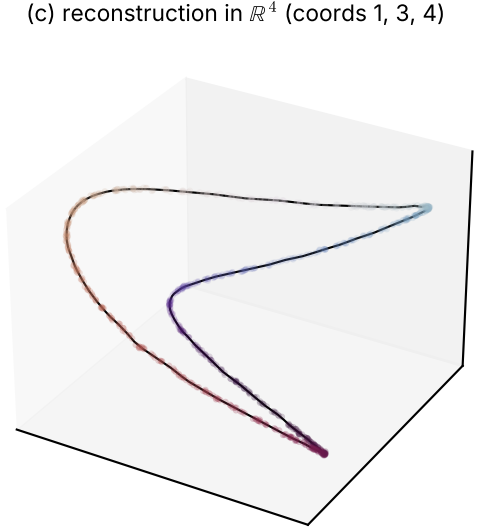
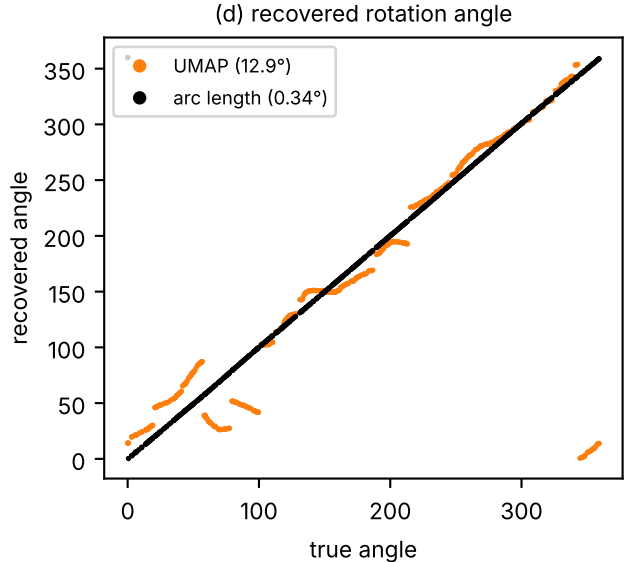

0°
60°
120°
180°
240°
300°

(a)


(b) PCA, two coordinates


(c) reconstruction in $\mathbb{R}^4$ (coords 1, 3, 4)


(d) recovered rotation angle
UMAP (12.9°)
arc length (0.34°)
recovered angle
true angle
0
50
100
150
200
250
300
350
0
100
200
300